# V-20—Printer requirements for volumetric liquid meters equipped with automatic temperature compensation (ATC)

**Date:** 2008-04-01

**Bulletin:** V-20 (rev. 1)

**Category:** Volume

**Document(s):** W&M Regs, section 129, 149, 257; W&M Specs SVM 1 section 27, 36 and SVM 2 section 14

**Supercedes:** V-20

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## 1.0 Purpose

The purpose of this bulletin is to outline the printing requirements for some typical liquid measuring devices that are equipped with automatic temperature compensation (ATC).

# 2.0 Scope

This bulletin applies to all fuel dispensers, refuellers, vehicle mounted and loading rack meters equipped with ATC.

# 3.0 Background

The original information contained within this bulletin was developed in 1996 as an internal guideline document for Measurement Canada Staff to follow. In an effort to satisfy requests from clients, the printing requirements for ATC devices were converted into bulletin format.

# 4.0 Policy

**4.1** Cashier consoles have not been checked by Measurement Canada's Engineering and Laboratory Services Directorate (ELSD) for many years now. In the past, the ELSD only evaluated the console if it had the ability to change the unit price on the dispenser. The console should be evaluated in the field for agreement with the primary means of registration on the device during a regular inspection if it has printing capabilities.

**4.2** For retail applications where the customer normally views the transaction, the statement "volume corrected to 15°C" is not required on the receipt/ticket if the primary register is so marked.

**4.3** A cash register that is separate and not interfaced to the measuring system (dispenser, console, card reader, etc.) does not have to comply with the printing requirements of section 129 of the *Weights and Measures Regulations* or section 27 of Ministerial Specification SVM-1.

**4.4** The printing requirements for fuel dispensers, refuellers, truck meters and loading racks are summarized in the table below.

# Printing requirements

| | Fuel Dispensers | Refuellers | Truck Meters | Loading Racks |
|---|---|---|---|---|
| **Net Volume** | ✔ A | ✔ A | ✔ | ✔ |
| **Gross Volume** | | | ✔ D | ✔ |
| **Volume Correction Statement** | | | ✔ | ✔ |
| **Density** | | | ✔ E | ✔ E |
| **Cub. Coefficient of Expansion** | | | | ✔ G |
| **I.D. # of Meter** | ✔ C | ✔ C | ✔ B | ✔ |
| **Unit of Measure** | ✔ A | ✔ A | ✔ A | ✔ A |
| **Monetary Unit ($ or ¢)** | Compulsory if a monetary value is mentioned | | | |
| **Unit Price** | Compulsory if the total price is indicated | | | |
| **Name and address of Trader** | ✔ A | ✔ A | ✔ A | ✔ A |
| **I.D. # of the Vehicle** | | | ✔ F | |
| **Address where meter is installed** | ✔ A | ✔ A | ✔ A | ✔ A |
| **Date of Transaction** | ✔ A | ✔ A | ✔ A | ✔ A |

A Also applies to auxiliary equipment such as card readers, dispenser control consoles, key systems, money acceptors (cash registers not included).

B Two meters on the vehicle—both the ticket and the meter must be identified with the I.D. number or product name; one meter/multi product—no I.D. number is required.

C Where more than one measuring device is located at a trader's premises, the I.D. number of the device is to be marked on the ticket or receipt.

D
i. For bulk wholesale applications only (where the purchaser is not the final user of the product), and includes the sale of aviation fuel (section 14 of SVM-2).
ii. Retail trade applications (where the purchaser is the final user of the product) such as, home heating oil, propane and anhydrous ammonia, are not required to print the gross / uncompensated volume on the meter ticket.

E Although the actual density value of the product is preferred, the product name is permissible when standard densities are used, i.e., (.510/propane, .730/gas, .840/diesel).

F Identification number of the vehicle is required if the address where the metering assembly is installed is not indicated on the ticket (SVM1,27(b)).

G Applicable to products where the VCF(s) are derived from API Table 54C (refer to Measurement Canada Bulletin V-10).

**4.5** Where one or more meters are installed in a system to blend products, the information required by the table in section 4.4 must be printed according to how the components of the blended product are measured.

**4.5.1** In the case where two or more components of the blended product are measured separately, each through an approved meter, the information required by this bulletin for each of the components must be stated and appropriate for the products.

**4.5.2** In the case where two or more components of the blended product are combined prior to being measured by an approved meter, the information required by this bulletin must be stated appropriately for that product blend. Information from a non-trade meter, such as in a side-stream blending system, must not appear on the ticket or be clearly identified as a not for use in trade volume.

**4.6** For all applications, the metering system's printing capabilities should be confirmed with the applicable Notice of Approval.

# 5.0 Revision

The purpose of revision 1 was to revise the bulletin format and sequence and to include a scope and background information; clarify requirements for non-petroleum and blended products; make minor editorial corrections and to update Measurement Canada's directorate.

**Date modified:**
2017-03-28